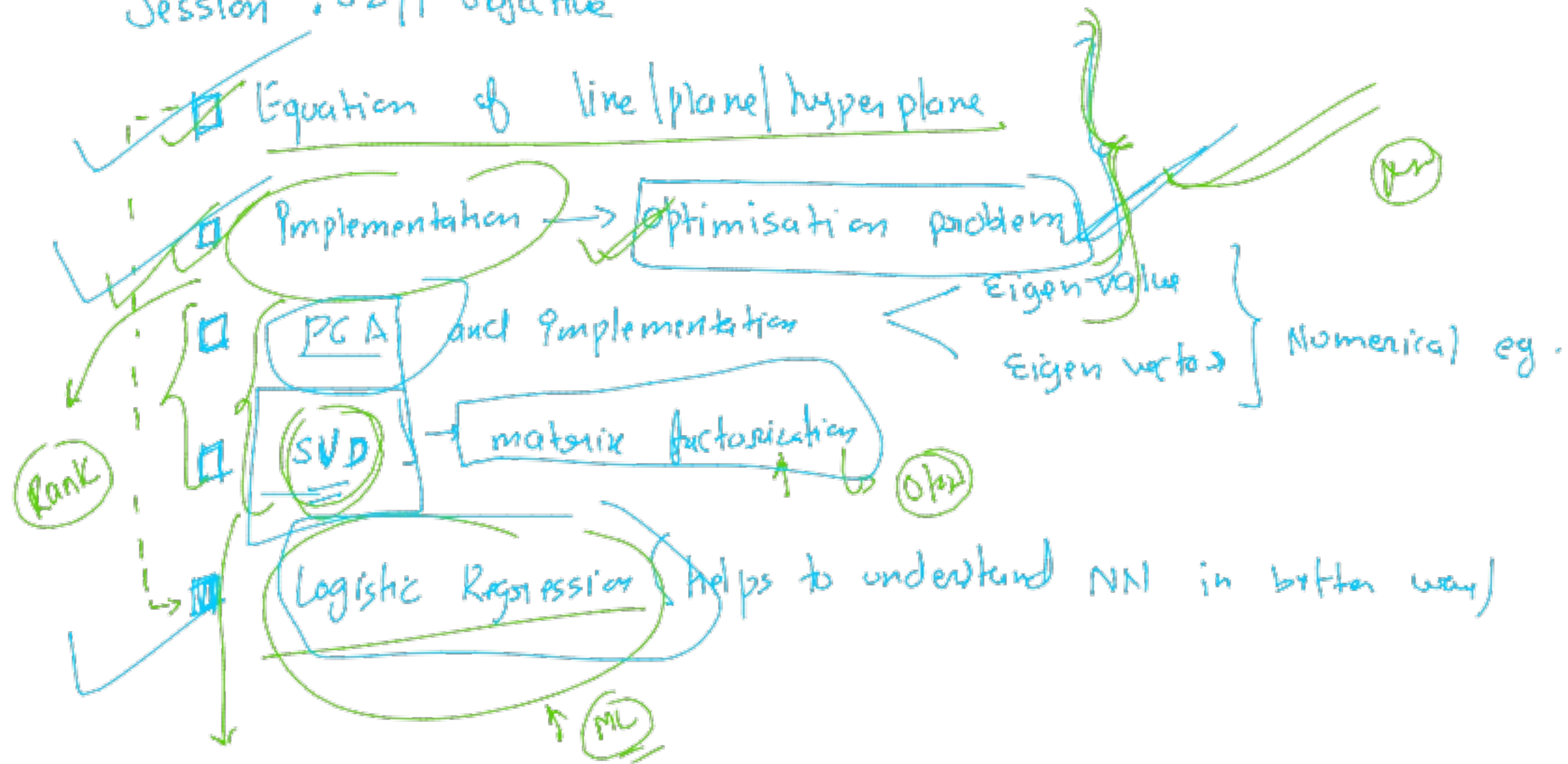


Session : 02 // Objective

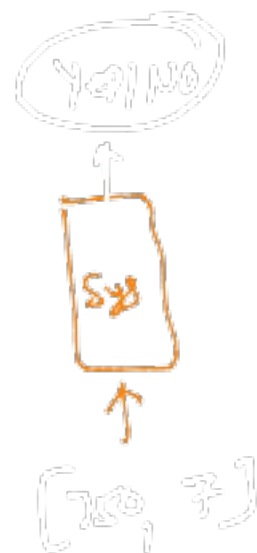


↑
n-custom

cibil-score	income	Status
[750	17]	Yes
[650	16]	Yes
[	]	:
:	:	:
:	:	:

matrix

$\begin{matrix} & cs' \\ \sin(&) \end{matrix}$
 $\begin{matrix} & i \\ \sin(&) \end{matrix}$

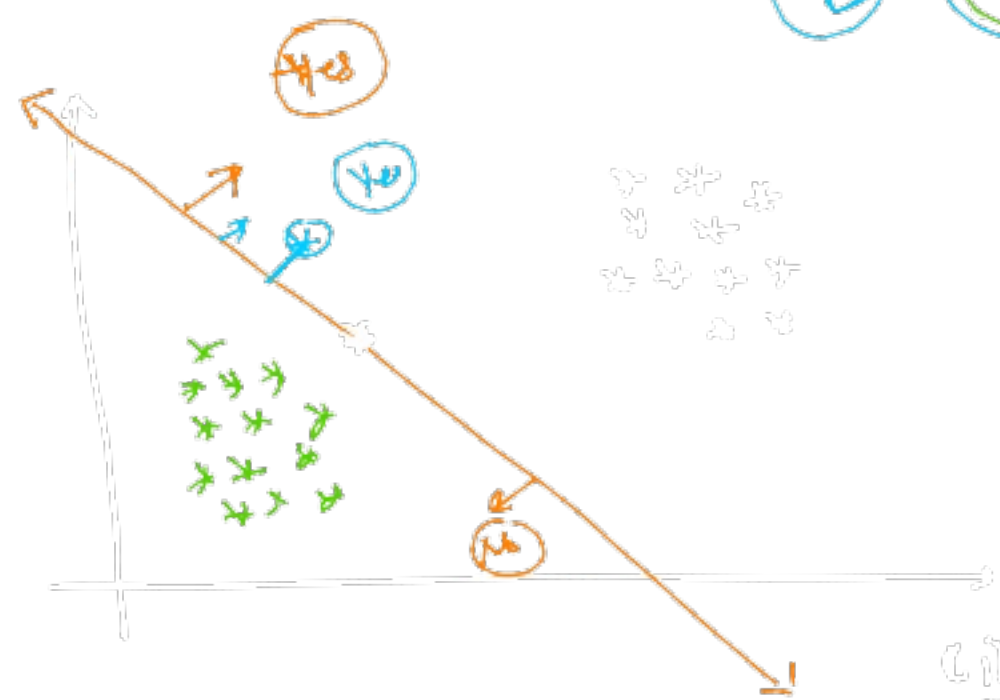


Handwritten signature or scribble.

① linear separable data

NPU

input



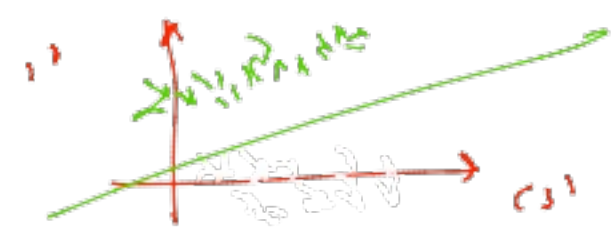
line $|x_1|$

② non-linear data



linear space

③ transformation



equation of line

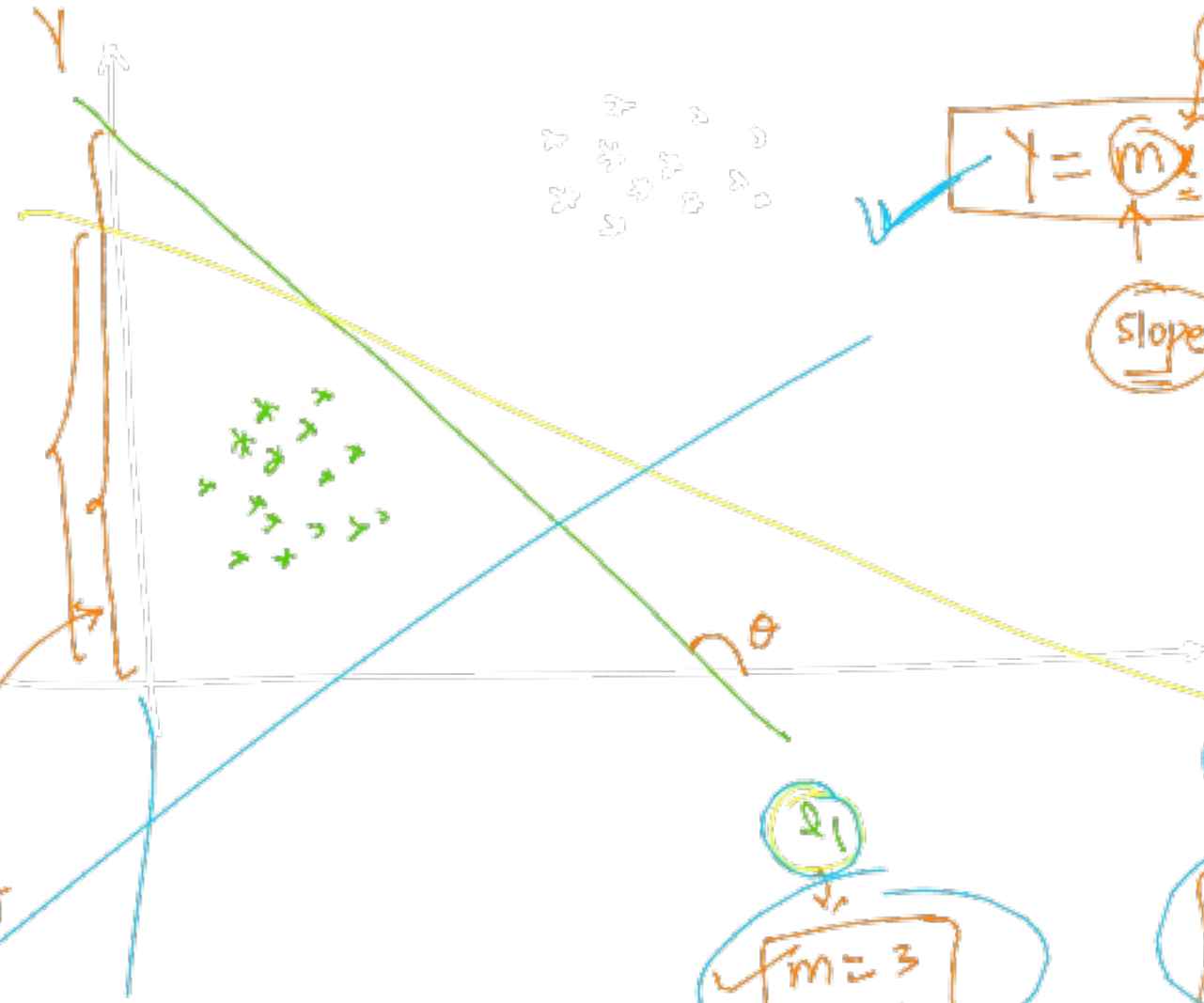
↓
general equation of line (form)

$$ax + by + c = 0$$

$$by = -ax - c$$

$$y = -\frac{a}{b}x - \frac{c}{b}$$

m
 y -intercept



(data)

$$y = \hat{m}x + c$$

Slope y -intercept

Q_1

$$m = 3$$
$$c = 4$$

Q_2

$$m_2 = 6$$
$$c = 8$$

$$Y = 3x + 4$$

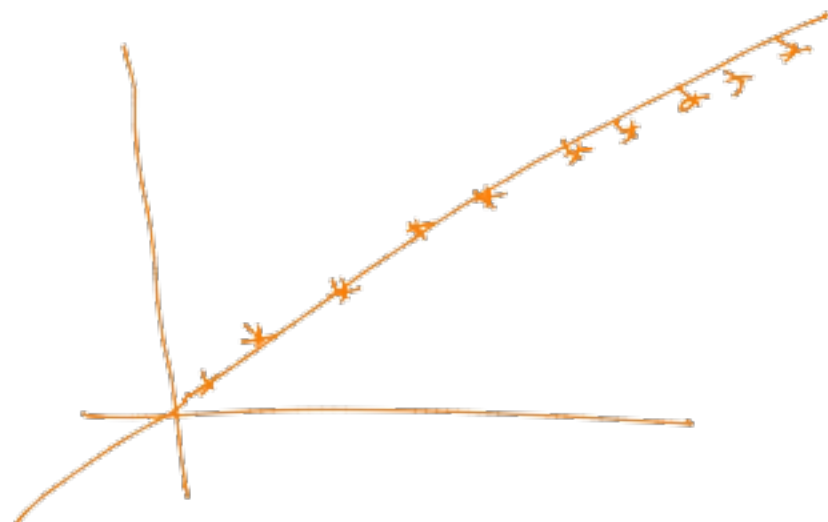
↑

↘

$$\left. \begin{array}{l} m = 3 \\ c = 4 \end{array} \right\}$$

$$Y = 13$$

7
5
4
6
9
1
1
1



find
out

$$3x + 4y + 3 = 0$$

equation of line

$$4y = -3x - 3$$

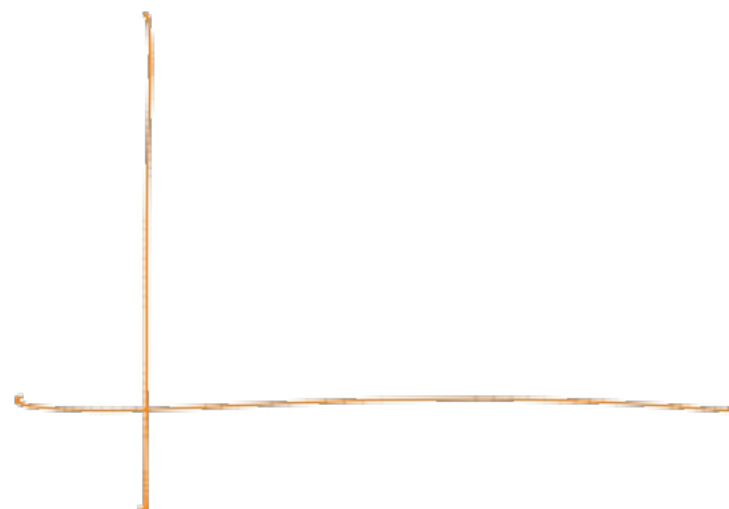
$$y = -\frac{3}{4}x - \frac{3}{4}$$

$$m = -\frac{3}{4}$$

↑

$$c = -\frac{3}{4}$$

↑

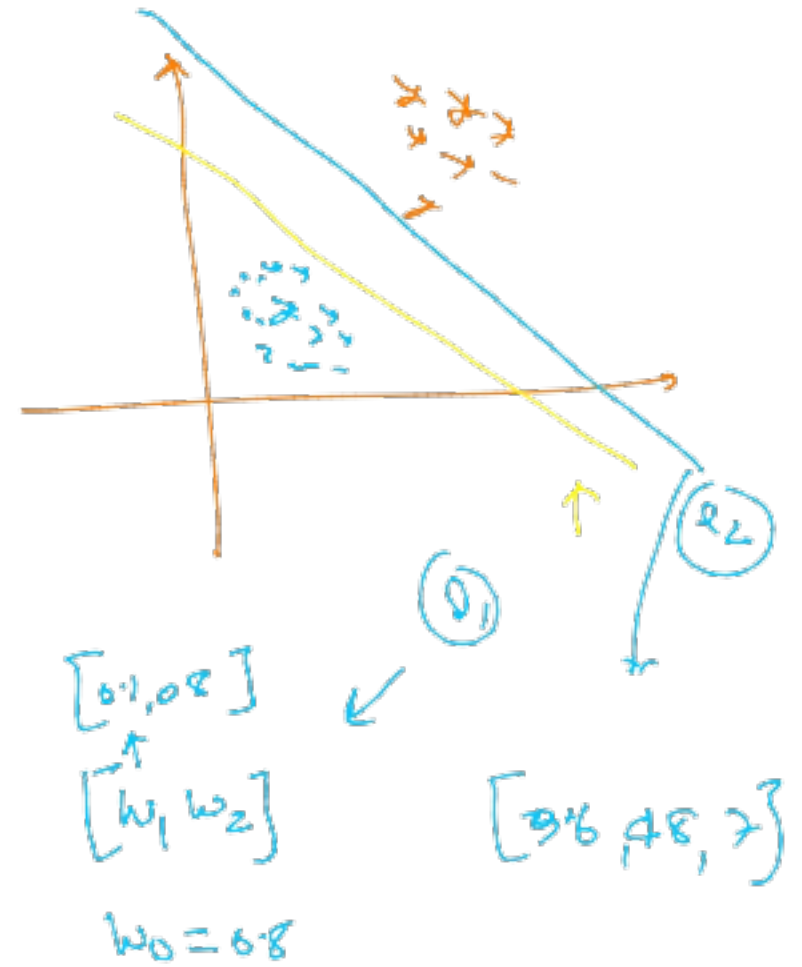


cibil-score income system

equation of line

$$w_1 \times \text{cibil-score} + w_2 \times \text{income} + w_0 = 0$$

w_1 w_2 w_0
 β β β
 36 48 7



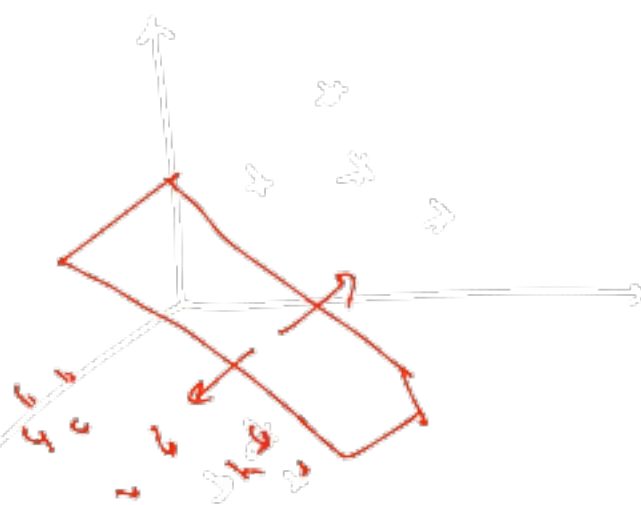
(2D) → line

(3D) → plane

$$w_1 * f_1 + w_2 * f_2 + w_3 * f_3 + w_0 = 0$$

ndim → Hyperplane

$$w_1 * f_1 + w_2 * f_2 + w_3 * f_3 + \dots + w_n * f_n + w_0 = 0$$



equation of line
↓

f_1	f_2	f_3	w
()	()	()	
()	()	()	
()	()	()	

$$w * f_1 + w_2 * f_2 + w_0 = 0$$

Derive

equation of line

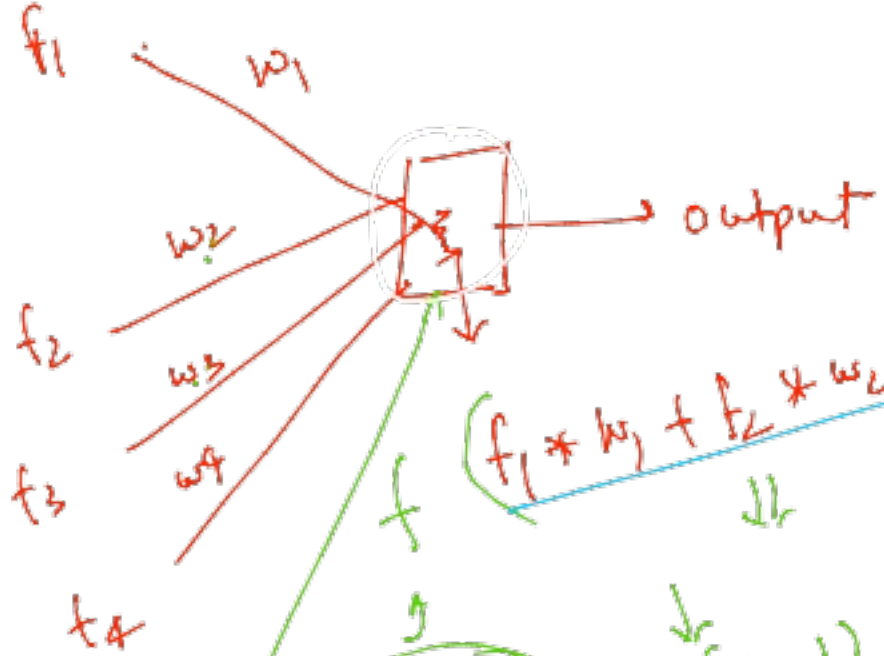
$$\underbrace{[w_1 \ w_2 \ w_3 \ \dots \ w_n]}_{\uparrow} \times \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ \vdots \\ f_n \end{bmatrix} + w_0 = 0$$

✓

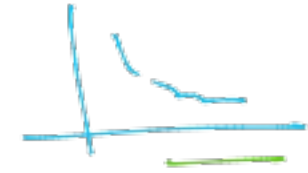
$$w_0 + \sum_{i=1}^n w_i^T \cdot x_i = 0$$

✓

$$\begin{bmatrix} \vdots \\ \vdots \end{bmatrix} \cdot \begin{bmatrix} \vdots \\ \vdots \end{bmatrix}$$



loss ↓



$$(f_1 * w_1 + f_2 * w_2 + f_3 * w_3 + w_0) \in 0$$

$$[w_1 \ w_2 \ w_3 \ w_4]$$

w_0

Activation Function

↓ (0-1)

(0.5)

Yes

No

Standard

$$[0.1, 0.2, 0.3, 0.1]$$

$$w_0 = 3$$

$\frac{1}{2}$

350

368

350

800

↓

Sum of

error

350

f

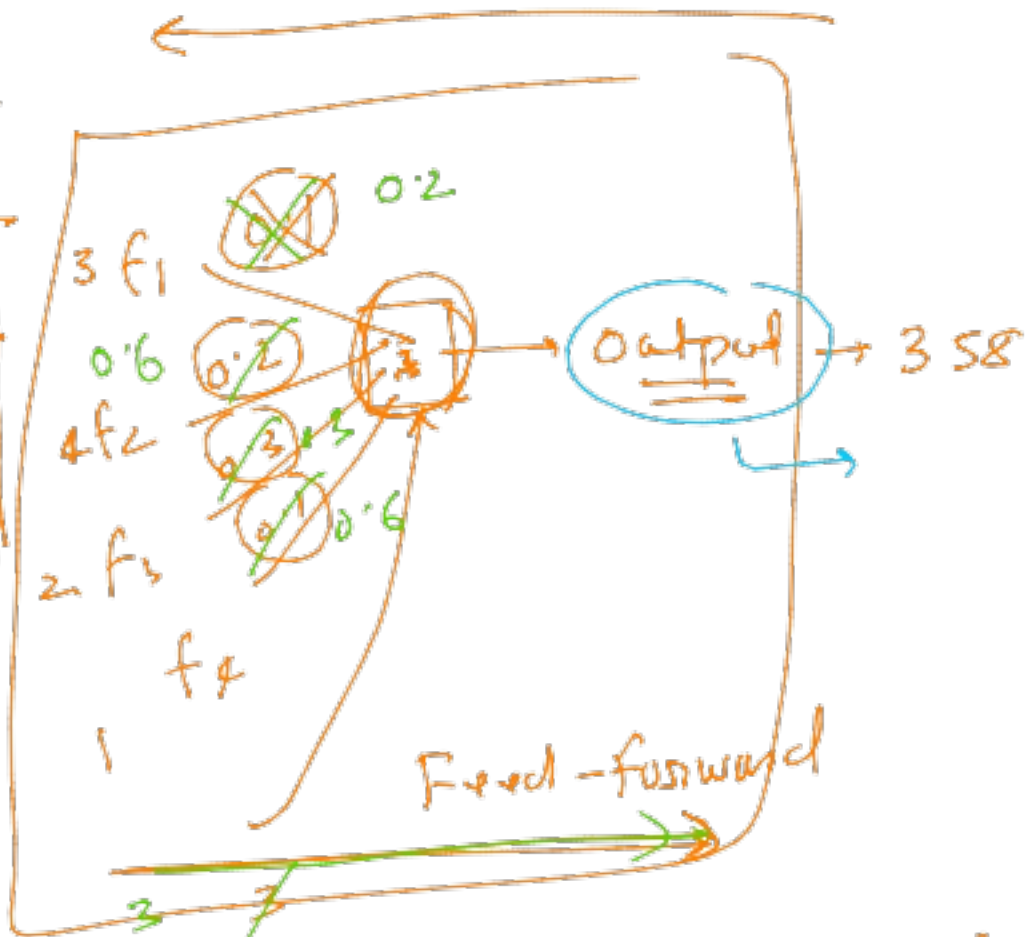
~~Sum~~

f_1	f_2	f_3	f_4	Sal-8	$\hat{y}$	error
3	4	2	1	340	358	$(18)^2$
4	5	6	7	369	400	31
7	8	1	2	359	450	100
9	8	1	2	754	460	500

Sum of error

4500

(ii)



$$(i) (3 \times 0.1) + (4 \times 0.2) + (2 \times 0.3) + (1 \times 0.4) = 3.58$$

Saw this weight

4 input

↓

(4 weights + 1 bias)

↓

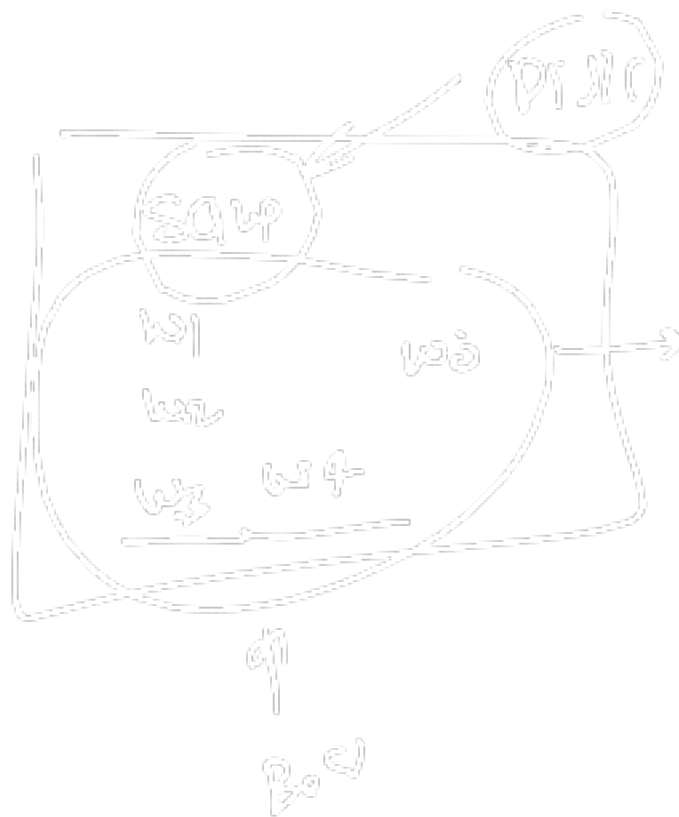
Σ

↓

vector → saw it local →

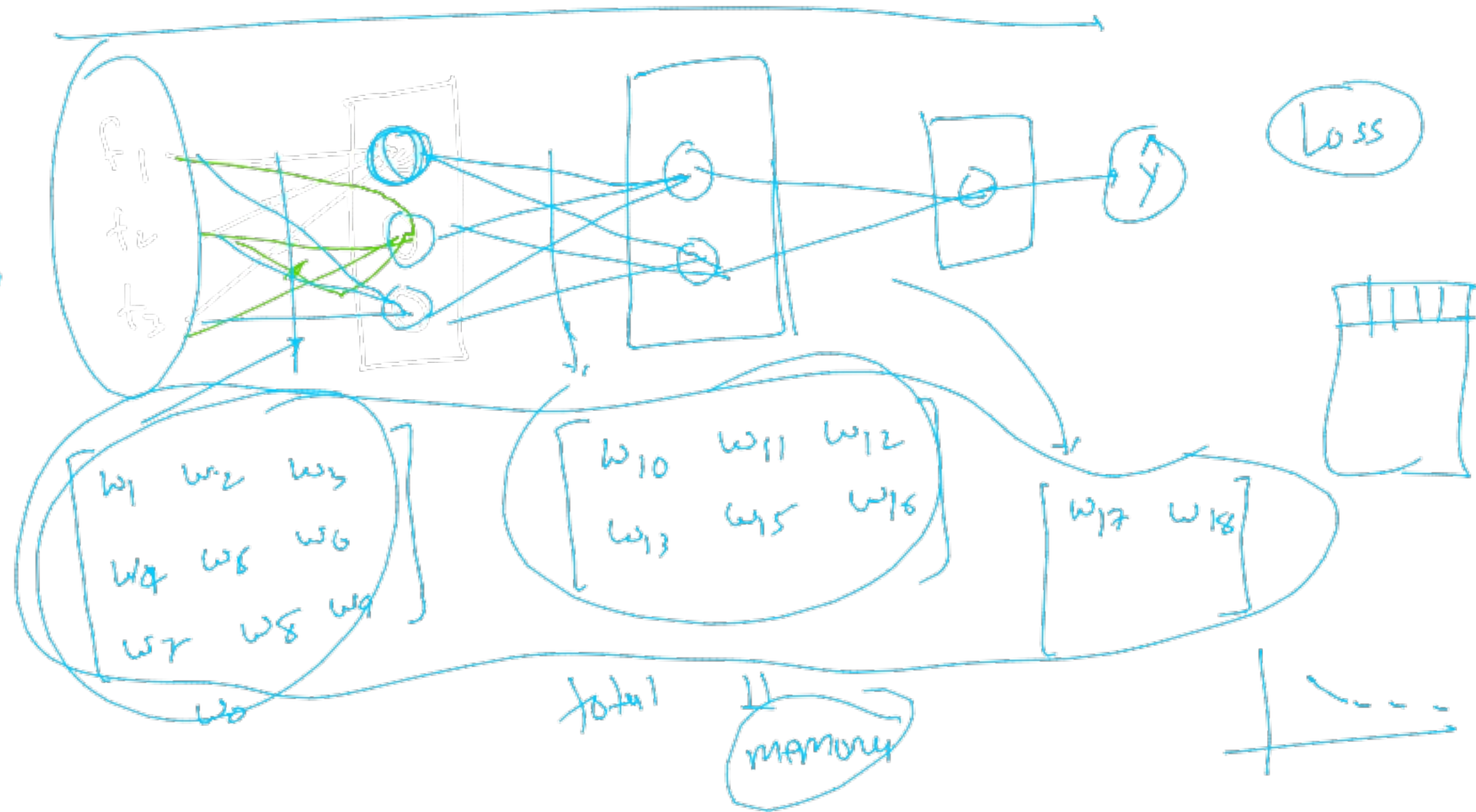
50 byte

Input
(features)
 f_1, f_2, f_3, f_4



Output

34



7B

10m

10

1000 layers

128

128 x 10

✓
64

128 x 64

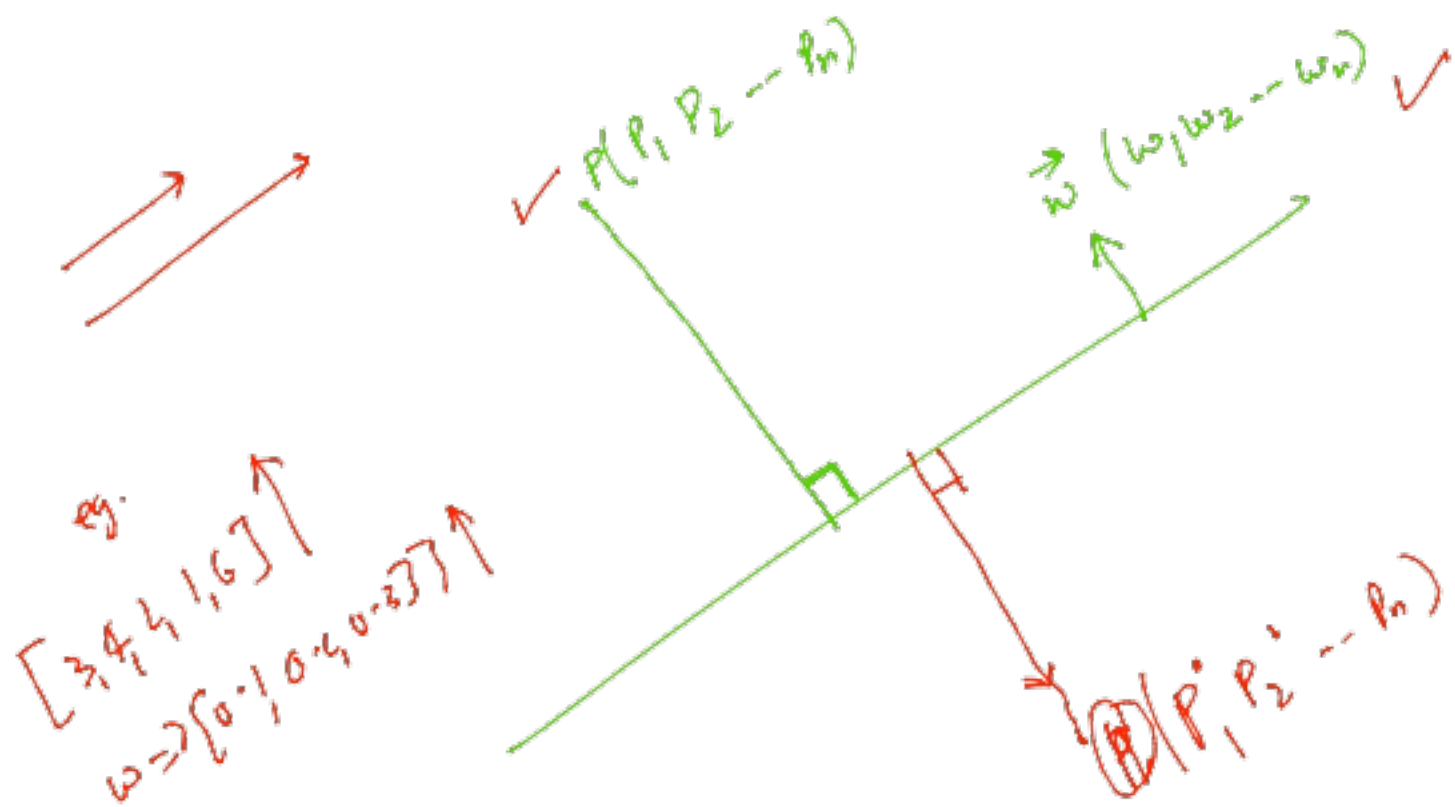
✓
32

✓
16

✓
8

✓
4

✓
2



$$d' = \frac{\vec{w}^T \cdot P}{\|\vec{w}\|} \Rightarrow \textcircled{+w}$$

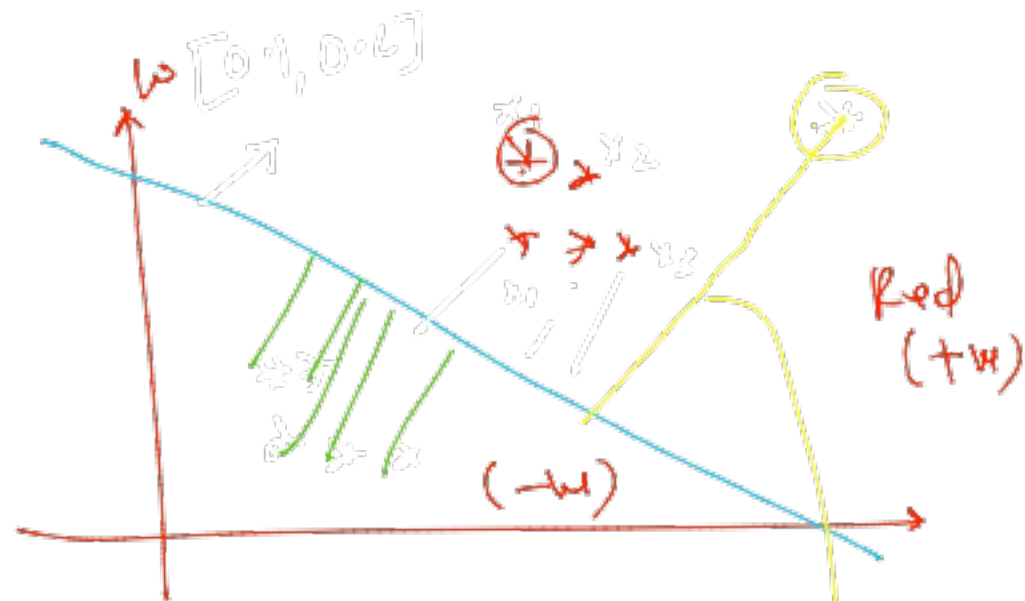
$$d'' = \frac{\vec{w}^T \cdot P'}{\|\vec{w}\|} \Rightarrow \textcircled{-w}$$

(signed distance)

$$\sum_{i=1}^n y_i w^T x_i > 0$$

Maximize (d)

ought directly

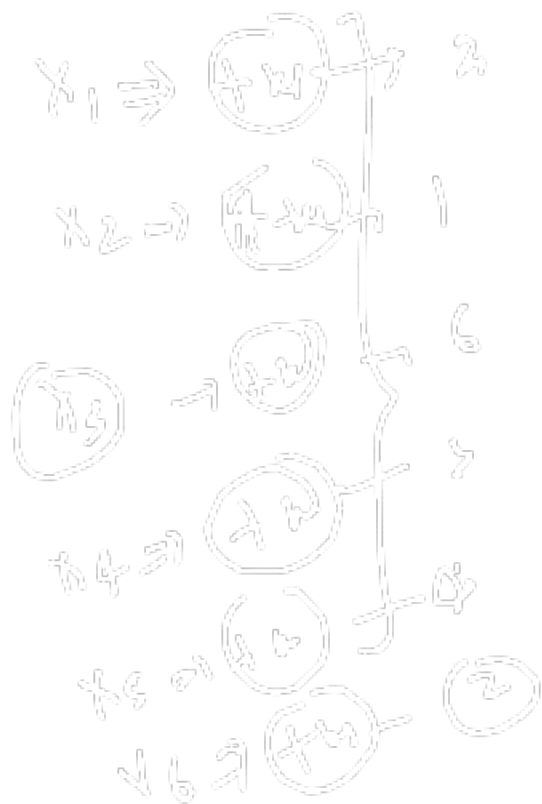


$$x_i \rightarrow \frac{w^T \cdot x_i}{(+)} * (+w) = (+w)$$

$$w_1 = 0.1$$

$$w_2 = 0.2$$

$$\frac{(-w)}{(-w)}$$



$$x_i \rightarrow \frac{w^T \cdot x_i}{(+)} * (+w) = (+w)$$

Classification Rule

✓

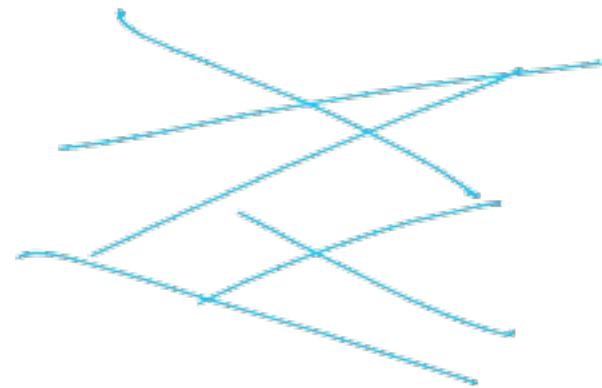
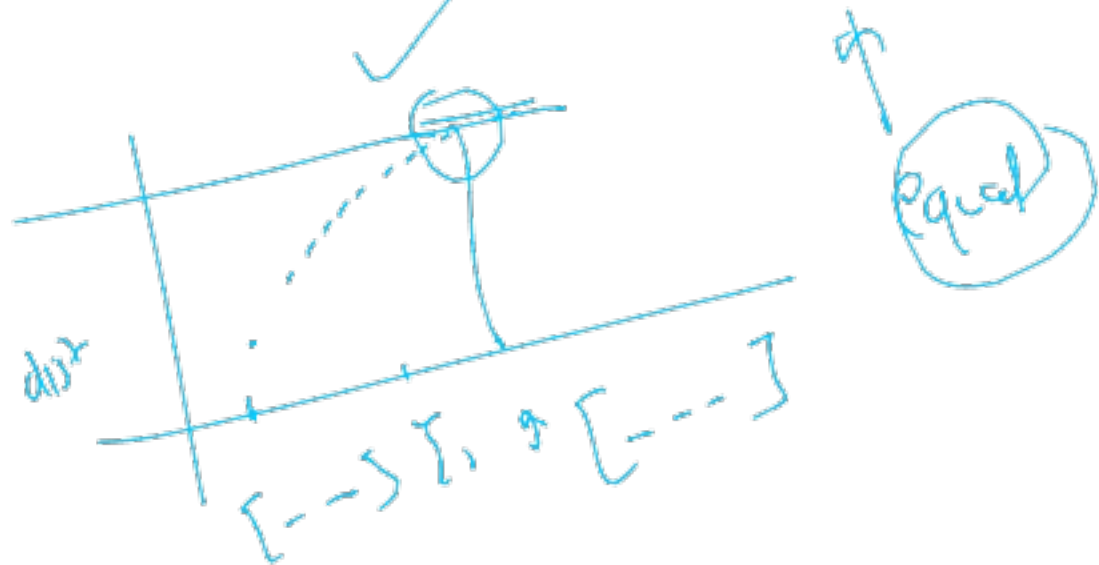
$$y_i * \underbrace{w^T \cdot x}_{\substack{\text{signed} \\ \text{distance}}} > 0$$

✓

for Bad fit line

$$\underset{w^k}{\text{argmin}} \sum_{i=1}^n (y_i w^T x_i)$$

when $y_i w^T x_i > 0$
for can



(+ve) (+ve)

$$3 + 4 + 2 + (-)(+10)$$

$\Downarrow$

✓ $(-1) + \underline{(-)(2)} + \underline{(-1)(-3)} + \underline{(-)(-4)} - \underline{(-1)(-2)}$

$$-1 + 11 \Rightarrow (10)$$

$\swarrow$ $w_1 \Rightarrow [0.2, 0.1]$



$$w_1 = [0.6, 0.8]$$

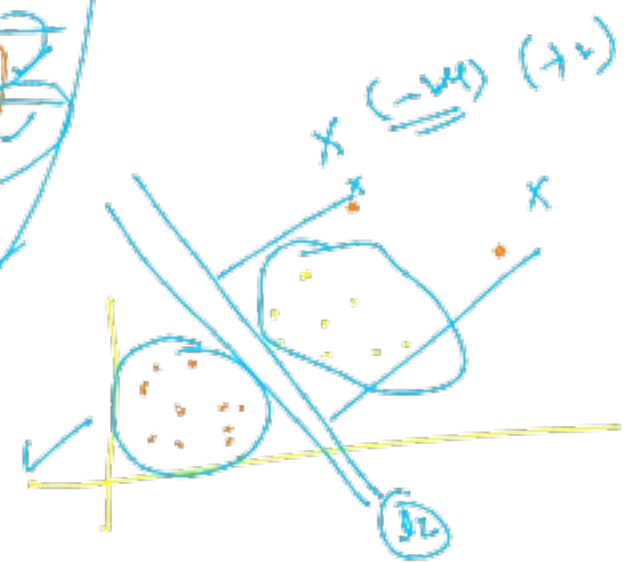
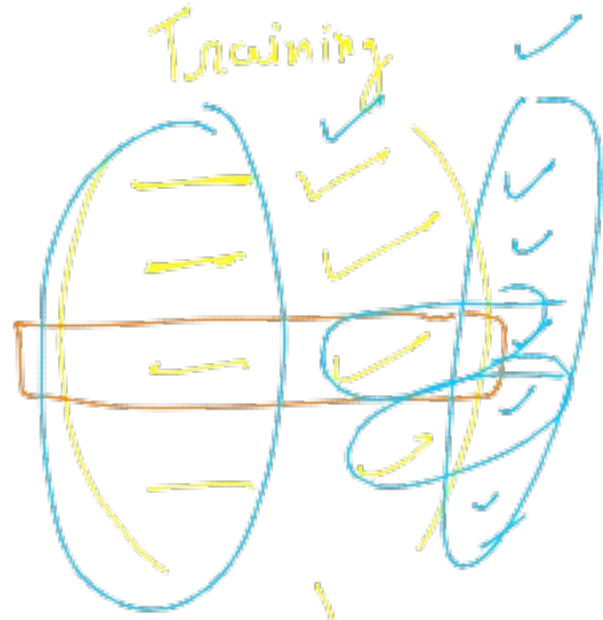
$$= (12) \checkmark$$

Data <

↔

Training

Testing



100%

PCA

Principle Component Analysis

10D dimensions

2D dimensions

compression

6 $\leftarrow$ $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$

8 x 16 x 1536

eigen value

Dimension Reduction Algorithm

Matrix factorization

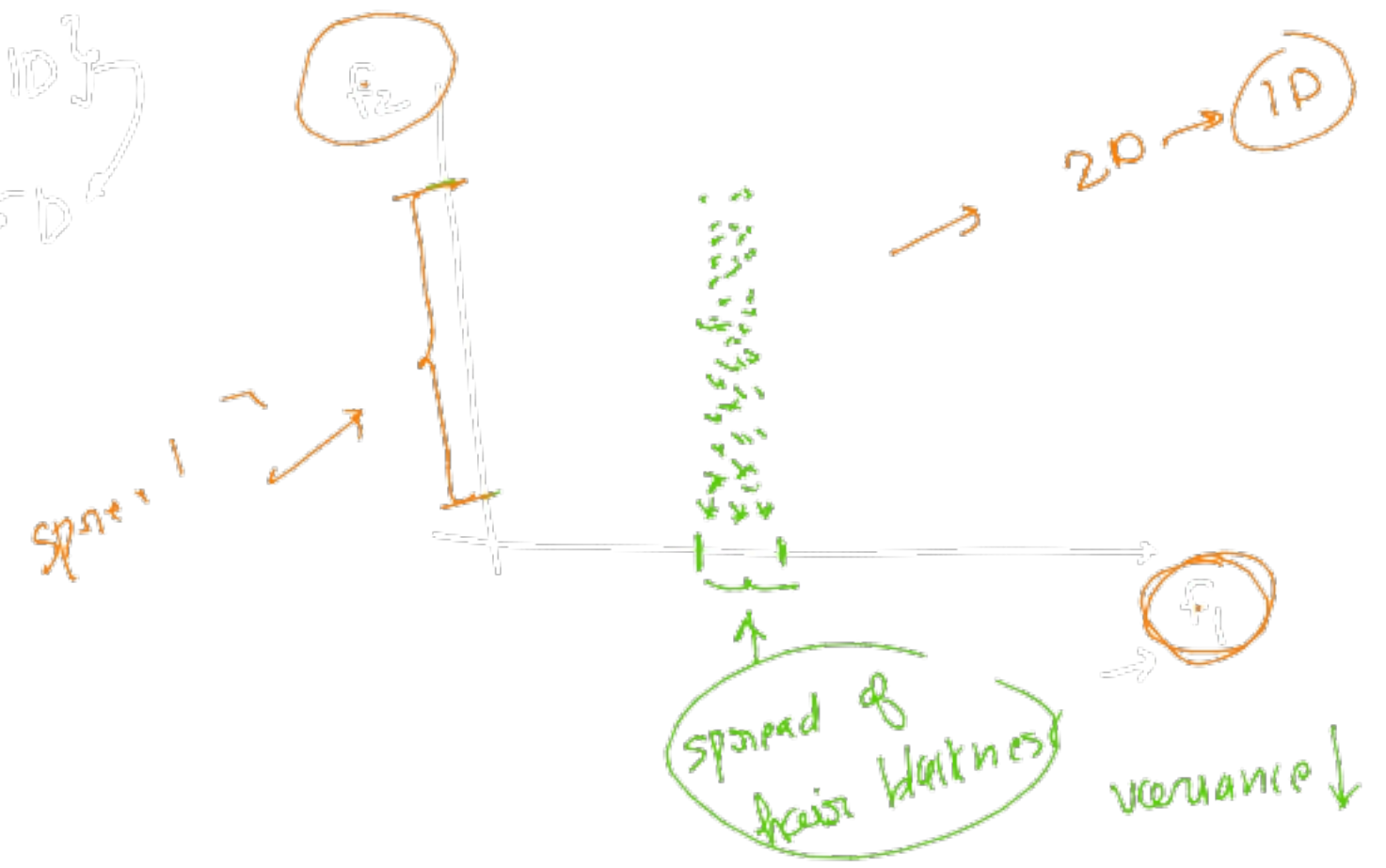
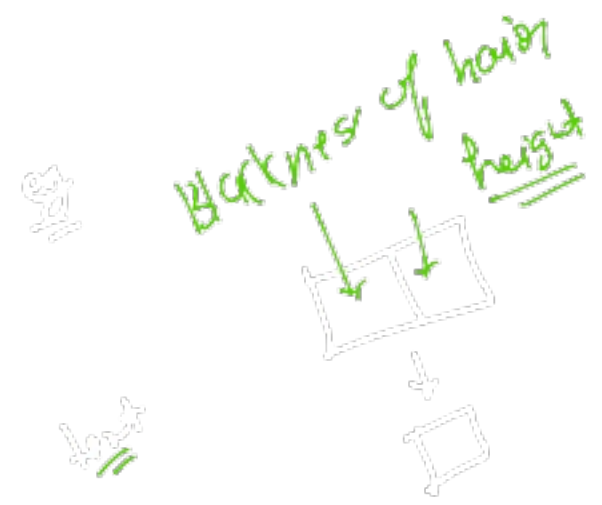
text $\rightarrow$

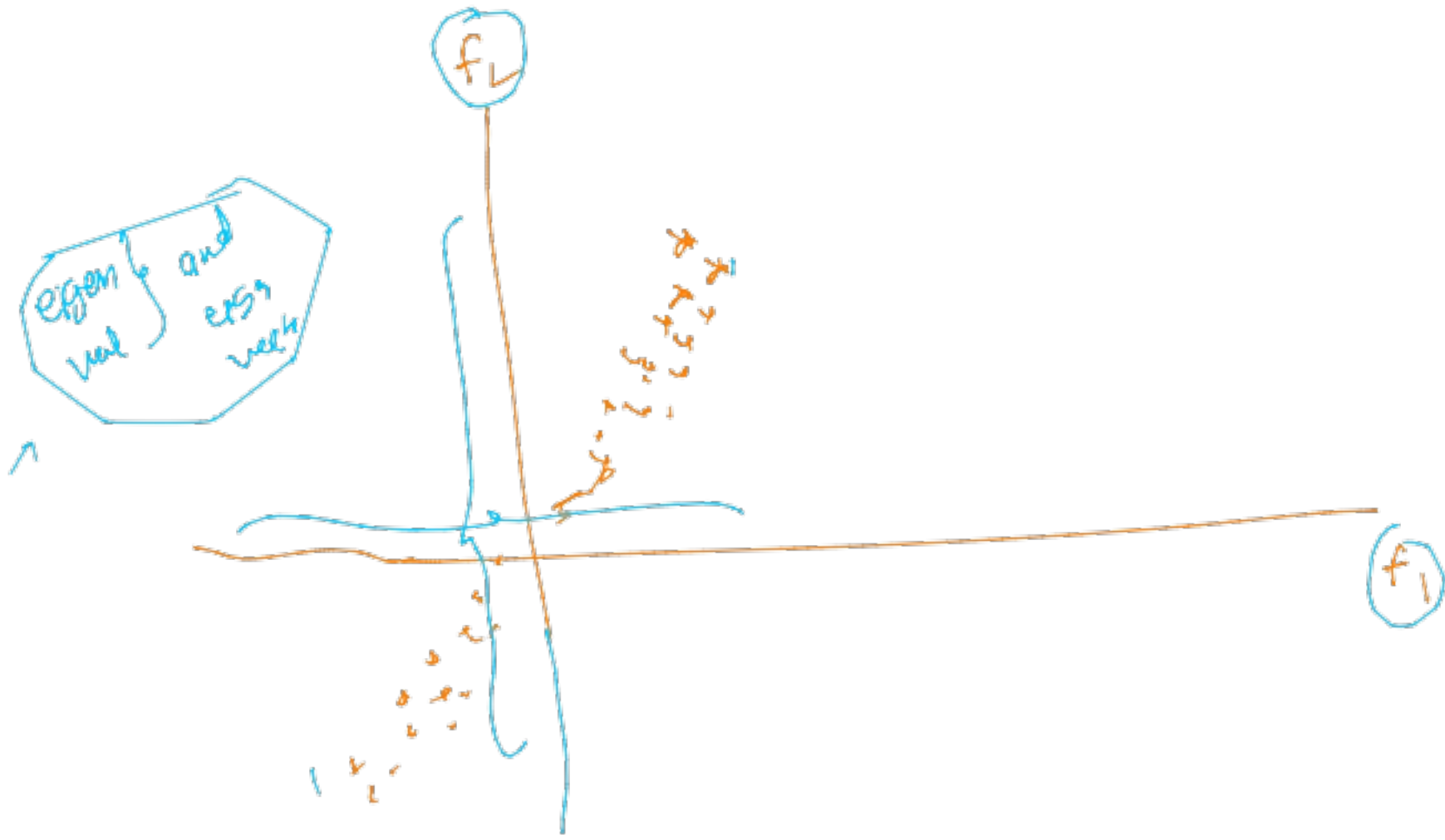
1					
---	--	--	--	--	--

k 1536

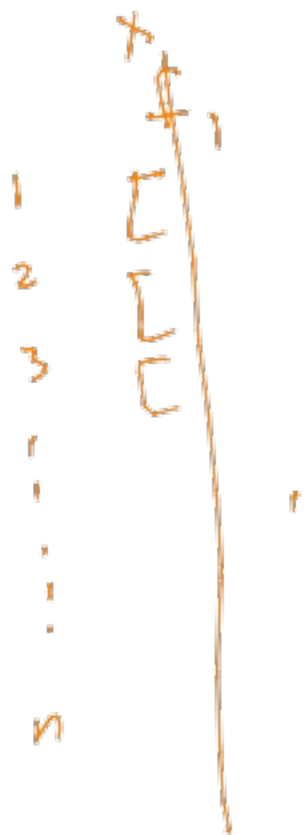
8 byte

$d \longrightarrow d'$ s.t. $d' < d$
 $\checkmark$ $2D \longrightarrow 1D$
 $10D \longrightarrow 5D$





X



f_2

